

Menlik Melesse

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Summary:

Data Scientist and Machine Learning Engineer with 6+ years of experience delivering advanced AI/ML and data-driven solutions across healthcare and enterprise domains. Expertise in predictive modeling, NLP pipelines, and end-to-end ML workflows, leveraging Python, R, SQL, Pandas, NumPy, Scikit-learn, Statsmodels, TensorFlow, PyTorch, XGBoost, LightGBM, MLflow, FastAPI, and Docker. Skilled in handling large-scale structured and unstructured datasets, implementing retrieval-augmented AI, and ensuring robust model performance and compliance. Adept at transforming complex data into actionable insights, improving patient outcomes, operational efficiency, and business impact.

Skills:

- Programming & Query Languages:** Python, R, SQL
- Machine Learning & Statistical Modeling:** Regression, Classification, Clustering, Feature Engineering, Model Evaluation & Validation, A/B Testing, Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, SHAP, LIME, Statsmodels, Pandas, NumPy
- NLP & Generative AI:** NLP Pipelines, Transformer Models, Hugging Face, LangChain, Retrieval-Augmented Generation (RAG), Vector Embeddings, FAISS, Pinecone, OpenAI APIs, Prompt Engineering
- MLOps & Model Deployment:** MLflow, Model Deployment (REST APIs), FastAPI, Docker, CI/CD for ML Pipelines, Model Performance Monitoring
- Data Engineering & Big Data:** PySpark, Apache Spark, Databricks, Airflow, Kafka, ETL/ELT Pipeline Development
- Cloud Platforms:** AWS (S3, Glue, Redshift, SageMaker), Azure (Azure ML, Azure Data Factory)
- Databases & Warehousing:** Snowflake, PostgreSQL, MySQL, MongoDB
- Healthcare Data & Compliance:** FHIR, HIPAA Compliance, Real-World Data (RWD), Data Governance
- Data Visualization & Reporting:** Tableau, Power BI, Plotly, Seaborn, Matplotlib, Advanced Excel
- Collaboration & Project Management:** Git, GitHub, JIRA, Confluence, Agile (Scrum, Kanban), Waterfall
- Core Competencies:** Problem Solving, Decision Making, Stakeholder Communication, Time Management, Cross-Functional Collaboration, Attention to Detail

Professional Experience:

Machine Learning Engineer, Cigna	Jul 2023 - Present USA
- Designed and deployed transformer-based NLP models using PyTorch, performing regression, classification, and feature engineering on 50,000+ structured and semi-structured healthcare records, improving annotation workflow throughput and model prediction reliability. - Built end-to-end ML pipelines with FastAPI and containerized using Docker, deployed via AWS SageMaker, reducing model retraining cycles by 2–3 days and enabling reproducible production workflows. - Developed RAG pipelines using LangChain and Hugging Face transformers, leveraging vector embeddings, FAISS, and Pinecone, accelerating unstructured clinical data retrieval and reducing manual annotation time by around 16 hours per week. - Processed semi-structured datasets from AWS S3 and MongoDB using PySpark and orchestrated via Airflow, performing ETL and feature engineering on 200+ features, improving data readiness for downstream ML tasks. - Applied XGBoost and LightGBM for predictive modeling, along with SHAP and LIME for interpretable AI, delivering actionable insights that reduced manual review overhead. - Integrated real-time Kafka streaming pipelines, performing Python-based analysis with Seaborn/Matplotlib and creating stakeholder dashboards in Tableau, enabling live monitoring of annotation KPIs, model outputs, and workflow efficiency for cross-functional teams. - Monitored deployed models using MLflow, ensuring ML workflow efficiency, deployment reliability, and SLA compliance for clinical NLP pipelines. - Collaborated cross-functionally with data engineers, data scientists, product managers, and clinical domain experts using GitHub, JIRA, and Confluence, applying Agile (Scrum/Kanban) to deliver high-impact ML solutions on schedule.	

- Developed regression and classification models using Scikit-learn and Statsmodels, segmenting over 2 million client records, improving predictive accuracy by 15% and informing business decisions.
- Automated ETL/ELT pipelines with Apache Spark and Airflow, transforming structured datasets stored in Snowflake and processing 500,000+ records daily, reducing ETL batch processing from 10 hours to 6–7 hours.
- Designed dashboards using Power BI, enabling stakeholders to track KPIs and model outputs for 10+ business units, improving reporting efficiency.
- Managed PostgreSQL and MySQL databases and deployed models via Azure ML, reducing model deployment time from days to under 24 hours and delivering 50+ production models.
- Conducted feature engineering and exploratory analysis using Pandas and NumPy, extracting predictive patterns and validating 200+ features for model development.
- Monitored deployed models using Python scripts, tracking metrics and pipeline efficiency, improving operational uptime by 18% across 50+ models weekly.
- Applied Model Performance Monitoring workflows for deployed predictive models in Azure ML, ensuring compliance with SLA metrics and operational reliability.

Education:

Bachelor of Science in Computer Science

Jackson State University, Jackson, MS, USA

Aug 2015 – May 2019